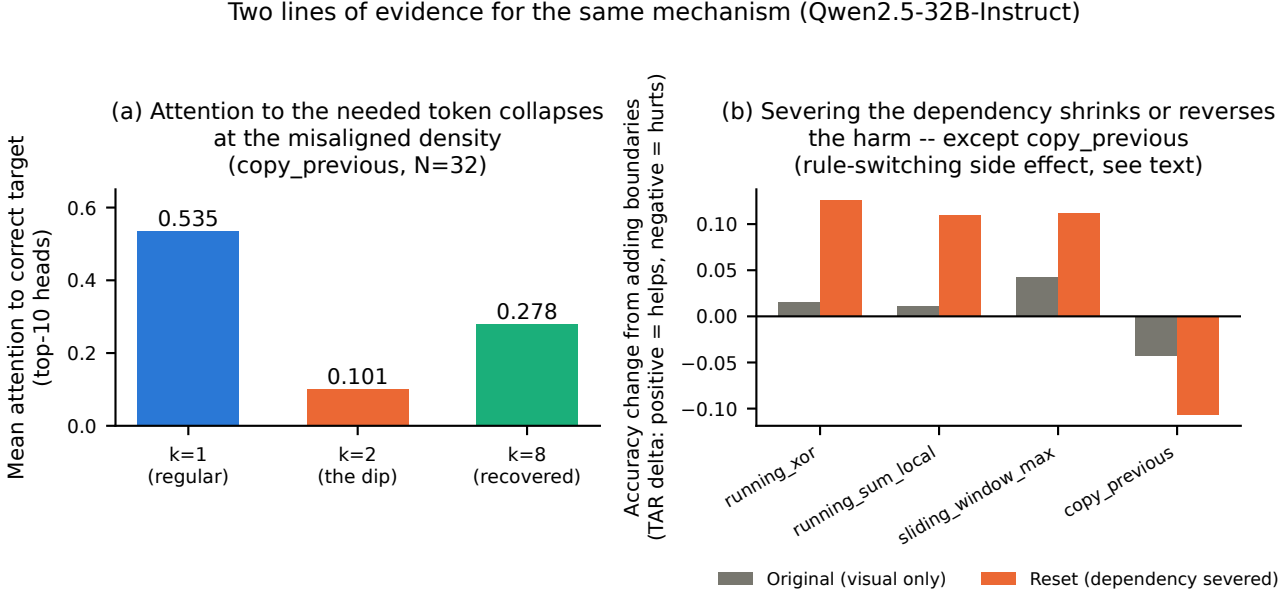

Two lines of evidence for the same mechanism (Qwen2.5-32B-Instruct)
(a) Attention to the needed token collapses at the misaligned density (copy_previous, N=32)
Mean attention to correct target (top-10 heads)
0.0
0.2
0.4
0.6
0.535
0.101
0.278
k=1 (regular)
k=2 (the dip)
k=8 (recovered)
(b) Severing the dependency shrinks or reverses the harm -- except copy_previous (rule-switching side effect, see text)
Accuracy change from adding boundaries (TAR delta: positive = helps, negative = hurts)
0.10
0.05
0.00
−0.05
−0.10
running_xor
running_sum_local
sliding_window_max
copy_previous
Original (visual only)
Reset (dependency severed)